DTU PCAS
Perception and Cognition
for Autonomous Systems

MSc Thesis Proposal

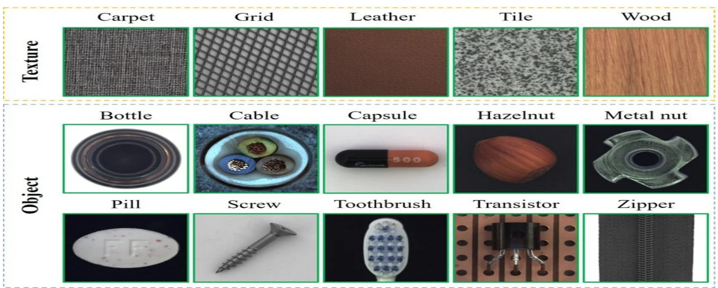
Re-Benchmarking Anomaly Detection: Classic Methods vs. Zero-Shot Foundation Models

Contact: Vasileios Tzouras (vatzo@dtu.dk)
Lazaros Nalpantidis (lanalpa@dtu.dk)

Motivation

Industrial anomaly detection is typically evaluated with classic embedding- or reconstruction-based methods (e.g., PatchCore, EfficientAD) that are trained per-class on normal-only images from benchmarks such as MVTec AD, MVTec LOCO, and VisA. Since 2025, a new generation of zero-shot detectors built on frozen vision foundation models (e.g., DINOv2/v3, CLIP-based backbones) has emerged, reporting state-of-the-art results without any per-class training data. These methods have mostly been benchmarked by their own authors against a narrow set of prior baselines, so it is unclear whether the established leader board rankings on MVTec AD/LOCO/VisA still hold once these newer zero-shot methods are evaluated under one consistent, independent protocol, and whether their advantage depends on the type of anomaly (textural/structural vs. logical) or on the amount of training data available.

Keywords: Anomaly Detection · Zero-Shot Learning · Vision Foundation Models · Industrial Inspection · Benchmarking



Problem Statement

Given the established industrial anomaly detection benchmarks (e.g., MVTec AD, MVTec LOCO, VisA), do current zero-shot foundation-model-based detectors match or outperform classic per-class-trained detectors under a single fair evaluation protocol and if the ranking changes, does it depend on anomaly type (structural vs. logical), on how much per-class training data is available to the classic methods, or on inference cost? Most existing comparisons fix one zero-shot method against a small, often outdated set of baselines; a systematic side-by-side re-benchmarking of both families does not yet exist.

Proposed Approach

The student will build a pipeline that runs two families of methods under same conditions: (i) classic per-class-trained baselines (e.g., PatchCore, EfficientAD) and (ii) recent zero-shot foundation-model detectors (e.g., AnomalyVFM, VisualAD, AVA-DINO), which are not yet in anomalib and will need to be integrated by the student from the authors' released code. All methods will be evaluated on MVTec AD, VisA, and MVTec LOCO using the same metrics. The student will then analyze failure modes per anomaly type, with MVTec LOCO's logical anomalies (wrong count, wrong arrangement) used specifically to test whether zero-shot methods, which were designed and tuned mainly on structural/textural defects, generalize to this harder anomaly category.

References to Get Started

[1] PatchCore: Towards Total Recall in Industrial Anomaly Detection

[2] EfficientAD: Accurate Visual Anomaly Detection at Millisecond-Level Latencies

[3] Beyond Dents and Scratches: MVTec LOCO AD – Logical Constraints in Unsupervised Anomaly Detection

[4] AnomalyVFM: Transforming Vision Foundation Models into Zero-Shot Anomaly Detectors

[5] VisualAD: Language-Free Zero-Shot Anomaly Detection via Vision Transformer

[6] Anomaly-Aware Vision-Language Adapters for Zero-Shot Anomaly Detection (AVA-DINO)

[7] anomalib: An Anomaly Detection Library